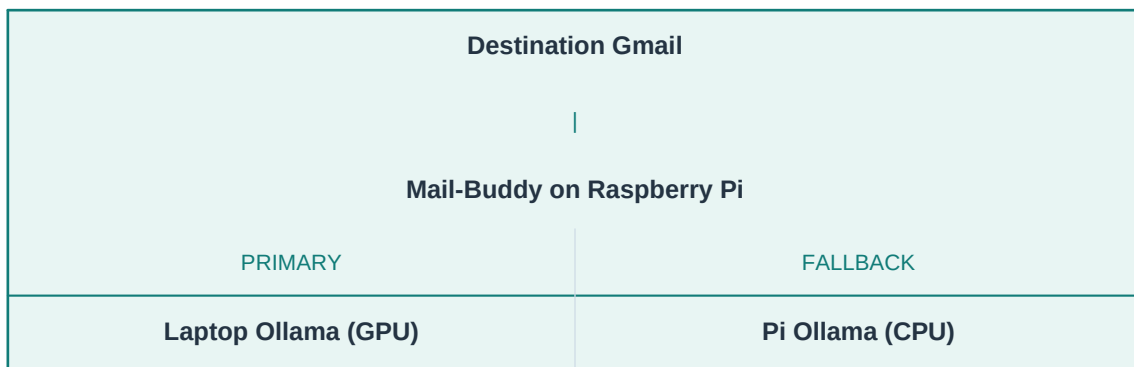


Raspberry Pi + Laptop GPU Personal Learning Deployment

Gmail integration, direct labels, failover, Pi learning, MLX QLoRA, rollback, and Home Assistant OS boundaries



Outcome

The laptop handles semantic model requests when reachable. If the request cannot connect or complete, Mail-Buddy retries the same request against the Pi. If both fail, unresolved mail remains in Inbox with Needs Review.

Direct Gmail labels

Labels are top-level names such as Security OTP, Bank Transactions, College Notices, Order Updates, and Needs Review. No Mail-Buddy/ prefix is created. Known legacy labels are renamed in place so their message membership is preserved.

Pinned model

```
llama3.2:3b-instruct-q4_K_M
```

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1. Security boundary

A redacted prompt derived from the message travels from the Pi to the laptop when the primary is enabled. The data remains on user-controlled devices, but it is no longer confined to the Pi. The prompt can still contain message text.

- Use only a trusted home LAN or a private Tailscale connection.
- Never expose TCP 11434 through router port forwarding.
- Restrict the laptop firewall so only the Pi can reach TCP 11434.
- Never configure a public Ollama host; the app rejects public addresses.
- Update the public privacy policy before enabling this option for users.

Accepted primary addresses

Accepted	10/8, 172.16/12, 192.168/16, Tailscale 100.64/10, IPv6 ULA fc00::/7
Rejected	Public IPs, DNS names, HTTPS URLs, URL credentials, query strings, fragments, paths

2. Prepare the laptop

Install Ollama natively. Apple Silicon Macs use Metal acceleration. Windows and Linux can use supported NVIDIA or AMD GPUs. Docker Desktop on macOS does not provide Ollama with GPU passthrough.

Pull and test the exact model

```
ollama pull llama3.2:3b-instruct-q4_K_M
ollama run llama3.2:3b-instruct-q4_K_M "Reply with READY only"
ollama ps
```

The Processor column in `ollama ps` reports GPU, CPU, or a split. Both laptop and Pi must retain the same pinned tag.

3. Assign a stable private address

Reserve the laptop LAN address in the router, such as 192.168.1.25, or install Tailscale on both devices and use the laptop's stable 100.64/10 address. Use a literal IP in Mail-Buddy; .local names are intentionally not accepted.

4. Expose Ollama to the Pi

Ollama binds to localhost by default. Change its listener and then enforce the source restriction with the laptop firewall.

macOS

```
launchctl setenv OLLAMA_HOST "0.0.0.0:11434"
```

Quit and reopen Ollama after setting the variable.

Linux systemd override

```
[Service]
Environment="OLLAMA_HOST=0.0.0.0:11434"

sudo systemctl daemon-reload
sudo systemctl restart ollama
```

Windows

Set the user environment variable OLLAMA_HOST to 0.0.0.0:11434, quit Ollama completely, and reopen it.

Connectivity check from the Pi

```
curl --fail --max-time 5 http://192.168.1.25:11434/api/tags
```

The response must list the pinned model. This test does not transmit mail content.

5. Configure Mail-Buddy on the Pi

Edit /opt/mail-buddy/.env. Replace the example address with the laptop's fixed private or Tailscale address.

```
MAIL_BUDDY_OLLAMA_PRIMARY_URL=http://192.168.1.25:11434
MAIL_BUDDY_OLLAMA_CONNECT_TIMEOUT_SECONDS=3
MAIL_BUDDY_OLLAMA_TIMEOUT_SECONDS=120
```

Do not replace the internal fallback URL. Compose supplies MAIL_BUDDY_OLLAMA_URL=http://ollama:11434 to the application. The Pi still downloads the model so fallback remains available.

Start and inspect

```
cd /opt/mail-buddy
./scripts/start-mail-buddy.sh
docker compose ps
docker compose logs --tail=100 app ollama model-init
```

6. How automatic failover works

1	Deterministic rules	Encrypted user rules and high-confidence deterministic evidence run first.
2	Laptop primary	Ambiguous mail is sent to the configured laptop Ollama endpoint.
3	Pi fallback	Connection, timeout, or HTTP failure triggers one retry against Pi Ollama.
4	Safe total failure	If both model requests fail, unresolved mail stays in Inbox with Needs Review.

7. Continuous personal learning

- Only owner-confirmed Needs Review corrections and answered accuracy questions become ground truth.
- The registry encrypts redacted sender domain, subject, and bounded Gmail snippet; complete bodies and attachment text are not retained for training.
- The Pi companion becomes eligible at 10 examples across at least two categories and uses deterministic sender-grouped five-fold validation.
- For ambiguous mail the companion advises Llama. Confident disagreement stays in Inbox with Needs Review; low confidence never overrides Llama.
- Disconnecting Gmail purges training examples, job/model metadata, and companion artifacts from the application database.

Companion readiness	10 confirmed examples; at least two represented categories
Companion promotion	5+ held-out predictions; accuracy $\geq 65\%$; no macro-F1 regression
QLoRA readiness	200 confirmed examples; 10+ per included category; sender-isolated train/valid/test
QLoRA promotion	Accuracy $\geq 70\%$ and no production, macro-F1, sensitive-recall, schema, injection, or failover regression

Schedule and dashboard

Automatic training defaults to every seven days at 02:00 in TZ (Asia/Kolkata by default). The dashboard offers 1, 3, 7, 14, or 30 days and displays readiness, category counts, last/next run, phase, metrics, failures, active versions, rejected versions, manual companion training, and rollback. Internal timestamps remain UTC.

8. Install the Apple Silicon MLX trainer

Use a 16 GB Apple Silicon Mac with native Ollama, FileVault, AC power, and at least 20 GiB free. Install `mlx-lm[train]` in a private virtual environment. Build `llama.cpp`, then pin both its exact commit and the SHA-256 of `llama-quantize` in the `mode-0600` trainer configuration.

```
ssh-keygen -t ed25519 -f ~/.ssh/mail_buddy_trainer_ed25519
# Copy the .pub file to the normal Raspberry Pi OS host, then:
sudo /opt/mail-buddy/scripts/install-pi-trainer-key.sh /tmp/mail_buddy_trainer_ed25519.pub
mkdir -p ~/.config/mail-buddy
cp scripts/mail-buddy-trainer.env.example ~/.config/mail-buddy/trainer.env
chmod 600 ~/.config/mail-buddy/trainer.env
./scripts/install-macos-trainer.sh
```

The `launchd` agent checks hourly and catches up after sleep. The forced SSH key cannot open a shell: it can request privacy-safe status, acquire a locked due export, transfer candidate files only into the model inbox, record verified installation, promote, fail, or roll back.

Fixed QLoRA recipe

- MLX-LM QLoRA; seed 42; batch 1; gradient accumulation 8.
- Eight trainable layers; rank 8; scale 20; maximum sequence 1024.
- Prompt masking and gradient checkpointing; approximately five dataset epochs.
- Fuse/dequantize to FP16 GGUF, then checksum-pinned `llama.cpp` Q4_K_M quantization exactly once.
- Evaluate candidate and current production end-to-end through Ollama before any activation.

9. Dual-host release and recovery

1	Export	Pi checks enabled, local schedule, 200-example readiness, and unique database lock.
2	Train	Only redacted JSONL crosses SSH; temporary files use a mode-0700 directory.
3	Evaluate	Candidate and production metrics plus application safety tests must pass.
4	Install	Identical GGUF checksum and versioned tag are recorded for laptop and Pi.
5	Activate	Registry promotion is atomic; any partial failure leaves production unchanged.
6	Recover	Dashboard rollback restores a previous promoted tag; two prior versions remain registered.

Home Assistant OS on Raspberry Pi 5

Do not install this Docker Compose stack, trainer SSH account, or Ollama directly into Home Assistant OS. Keep HA OS as the appliance and run Mail-Buddy on a Mac, second Raspberry Pi OS host, or another supported Linux Docker host. Do not mount the Home Assistant Docker socket or disable app protection.

Privacy disclosure

Disclose encrypted redacted training excerpts and optional processing on a user-controlled laptop. Plaintext JSONL exists transiently during training. On SSD/APFS, deletion cannot guarantee physical erasure because of wear levelling; keep FileVault enabled. GGUF artifacts outside the application database must be removed separately during device decommissioning.

Operator diagnostics

```
docker compose exec -T app mail-buddy training-status
docker compose exec -T app mail-buddy recover-stale-training --older-than-hours 12
docker compose exec -T app mail-buddy rollback-main-model
launchctl print gui/$(id -u)/com.mail-buddy.trainer
```

10. Acceptance tests

Use only a dedicated Gmail test account and harmless synthetic messages.

Laptop primary

Keep laptop Ollama running, send an ambiguous synthetic message, confirm laptop Ollama logs or ollama ps show activity, and verify the Gmail label.

Pi fallback

Quit laptop Ollama, send a different synthetic message, inspect docker compose logs --since=5m app ollama, and confirm classification succeeds.

Both unavailable

Stop both Ollama instances. Confirm unresolved mail stays in Inbox and receives Needs Review.

11. Routine operations

Start Pi stack	<code>./scripts/start-mail-buddy.sh</code>
Stop safely	<code>./scripts/stop-mail-buddy.sh</code>
Container status	<code>docker compose ps</code>
Recent inference logs	<code>docker compose logs --since=5m app ollama</code>
Laptop processor	<code>ollama ps</code>
Disable primary	Blank <code>MAIL_BUDDY_OLLAMA_PRIMARY_URL</code> and restart

12. Troubleshooting

Laptop gets no requests	Run the Pi curl test; check <code>OLLAMA_HOST</code> , firewall, address, and exact model tag.
Primary URL rejected	Use <code>http://PRIVATE-IP:11434</code> with no path, credentials, query, or hostname.
Laptop uses CPU	Use native Ollama and verify GPU support and drivers with <code>ollama ps</code> .
Fallback is slow	Keep connect timeout at 3 seconds and use a stable LAN or Tailscale address.
Both models unavailable	Restore either service. Needs Review messages remain in Inbox for correction.

13. Go-live checklist

- ☐ Both machines list `llama3.2:3b-instruct-q4_K_M`.
- ☐ Laptop has a stable private or Tailscale address.
- ☐ Laptop firewall allows TCP 11434 only from the Pi.
- ☐ No router port forwarding exposes Ollama.
- ☐ Laptop-primary and Pi-fallback synthetic tests both pass.
- ☐ Both-hosts-down test keeps unresolved mail in Needs Review.
- ☐ Owner answers, not unanswered predictions, are the only training truth.
- ☐ Mac trainer refuses battery power, low disk, or an unpinned quantizer.
- ☐ Candidate checksum/tag is identical on laptop and Pi before promotion.
- ☐ Rollback restores the previous fine-tuned version.
- ☐ Home Assistant OS remains unchanged on its dedicated Pi.
- ☐ Public privacy policy discloses optional trusted-laptop inference.

Official references

Ollama FAQ and network configuration

<https://docs.ollama.com/faq>

Ollama hardware and GPU support

<https://docs.ollama.com/gpu>

Ollama macOS support

<https://docs.ollama.com/macos>

Ollama API introduction

<https://docs.ollama.com/api/introduction>

MLX-LM LoRA and QLoRA

https://github.com/ml-explore/mlx-lm/blob/main/mlx_lm/LORA.md

Ollama GGUF import

<https://github.com/ollama/ollama/blob/main/docs/import.md>